Adding TypeScript

At this point, we have a fully functional Rails 7 setup to start writing front-end code. We could stop here. However, instead, we're going to add TypeScript to our mix.

There's good news and bad news regarding using TypeScript with esbuild.

The good news is that esbuild will convert .ts files from TypeScript to JavaScript without any configuration changes on our part. The bad news is that all esbuild does is remove TypeScript type annotations and whatever other TypeScript-only stuff might exist in the file. Specifically, esbuild does not run the TypeScript compiler to determine if the code is type-safe. Because we want to use TypeScript specifically to determine if our code is type-safe, this seems less than ideal.

What we want to do is introduce the TypeScript compiler to our watched build process. The esbuild documentation says "you will still need to run tsc -noEmit in parallel with esbuild to check types," but there's no guidance on how to do that.

Enter the tsc-watch[29] package, which runs the TypeScript compiler in watch mode and allows us to specify what to do on success and on failure.

Let's install it:

```
$ yarn add --dev typescript tsc-watch
$ yarn add --dev @typescript-eslint/parser
$ yarn add --dev @typescript-eslint/eslint-plugin
```

This gives us TypeScript itself, tsc-watch, and a couple of plugins for the ESLint[30] linter. (I snuck ESLint into the code; we'll take a look at that later.)

TypeScript uses a configuration file at tsconfig.json. Here's the one we'll start with:

chapter_01/03/tsconfig.json

```
{
```